IT5009 MULTIMEDIA TECHNOLOGIES

ASSIGNMENT - III

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Explain the detailed characteristics of MDBMS. Discuss its key features and how it differs from a traditional database management system.

Multimedia Database Management Systems, or MMDBMS or MDBMS, are specialised systems designed to manage multimedia data such as

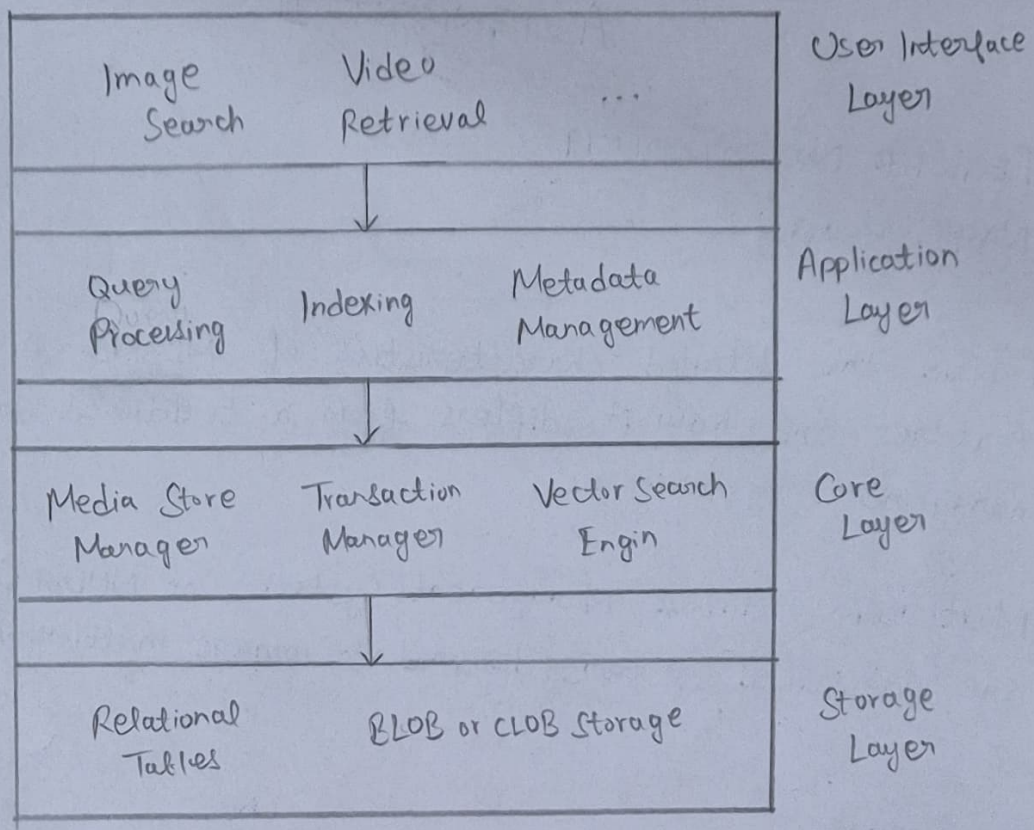
- → images
- → audio
- → video
- → text
- → graphics
- → animation

Unlike traditional databases, MDBMS handles data that is complex in structure, size, and retrieval requirements. These systems provide support for

- → multimedia content storage
- → indexing
- → querying
- → retrieval

and are used in a wide range of applications such as digital libraries, medical imaging, and multimedia content delivery.

ARCHITECTURE



Database : Stores multimedia data in various formats.

Media Storage Manager: Manages physical storage.

Query Processor : Processes user queries involving CBR and metadata filtering

Index Manager : Handles indexing based on attributes such as colour, texture, shape, and temporal characteristics.

User Interface: Allows users to interact with the system for querying and browsing

Transaction Manager: Ensures concurrency control, recovery, and transaction management.

CHARACTERISTICS AND KEY FEATURES

Data Types : → BLOB → Spatial
→ CLOB → Videos
→ JSON/XML → Audio

Data Modelling: → Structural Model
→ Temporal Model
→ Spatial Model

Metadata: → Data about data
→ Describes properties of multimedia files
→ Easy search, retrieval, and management
→ Descriptive
Structural
Administrative
Technical

Querying: → Uses a mix of SQL, NoSQL, and specialised query techniques.
→ SQL
NoSQL
CBR
MQL → Multimedia Query Languages

Transactions: They have the following characteristics
→ Long Duration
→ Real-Time Constraints
→ Heterogenous Media Types
→ High Volume of Data
→ Concurrency Control

BASE : → Basically Available
→ ~~Availability~~ Soft State
→ Eventual Consistency

Traditional Database	Multimedia Database
Structured format with tables, rows, and columns	Designed to store and manage multimedia data types
Data is organised in relations with predefined schemas.	Data may not have a strict schema, and can be unstructured/semistructured.
Each table represents an entity, and relationships between tables are defined through foreign keys.	Uses specialised formats to handle BLOBs.
Primary handles text and multimedia data.	Focuses on handling complex data types.
Supports integers, floats, strings, and dates.	Supports various multimedia formats and metadata.
Uses SQL.	Uses SQL extensions or specialised query languages designed for multimedia.
Designed for structured data operations like SELECT, INSERT, ...	Queries can involve searching for patterns, using techniques like CBR.
Optimised for transactions and operations on structured data.	Optimised for handling large volumes of multimedia data and retrieval.
Indexing strategies are focused on improving SQL query performance.	Uses techniques like indexing or audio signatures to improve search performance.
Used in applications requiring structured data management such as → financial systems → customer relationship management → enterprise resource planning	Used in applications that require management of multimedia content, such as → digital asset management → video streaming services → photo storage systems

You are tasked with designing a multimedia audio player for a mobile application. In this system, a producer thread decodes audio samples and places them into a shared buffer, while a consumer thread plays the samples through the device's speakers.

- The buffer can hold up to 8 audio samples.
- The producer takes 80 ms to decode each sample.
- The consumer takes 120 ms to play each sample.
- The buffer starts empty at time zero.

(i) How long will it take for the buffer to become full the first time?

Producer Speed PS = 80 ms

Consumer Speed CS = 120 ms

Initial buffer State → empty

In 1000 ms, producer decodes 12.5 samples.

consumer removes 1000/120 = 8.33 samples

So, net buffer gain = ~~12-8.7~~

12.5 - 8.33

= 4.17 samples per second

The buffer can hold upto 8 samples.

Time taken to fill buffer = $\frac{8}{4.17}$

= 1.918 s

(ii) Will the consumer cause a buffer underflow?

No, Since the producer is faster, the buffer will only keep filling up. The consumer will NOT cause a buffer underflow.

Write down when a buffer underflow does occur.

Let PS = 80 ms
CS = 70 ms

Now the consumer is faster than the producer.

Net buffer gain = $\frac{1000}{80} - \frac{1000}{70}$

$= 12.5 - 14.29$

$= -1.78$

Negative value of net buffer gain implies a buffer underflow.

Define memory allocation and buffering with a real-time example of each.

Memory Allocation: Process of reserving portions of RAM to store multimedia data temporarily during execution. Ensures applications have enough space to hold working data like video frames, audio samples, or image layers.

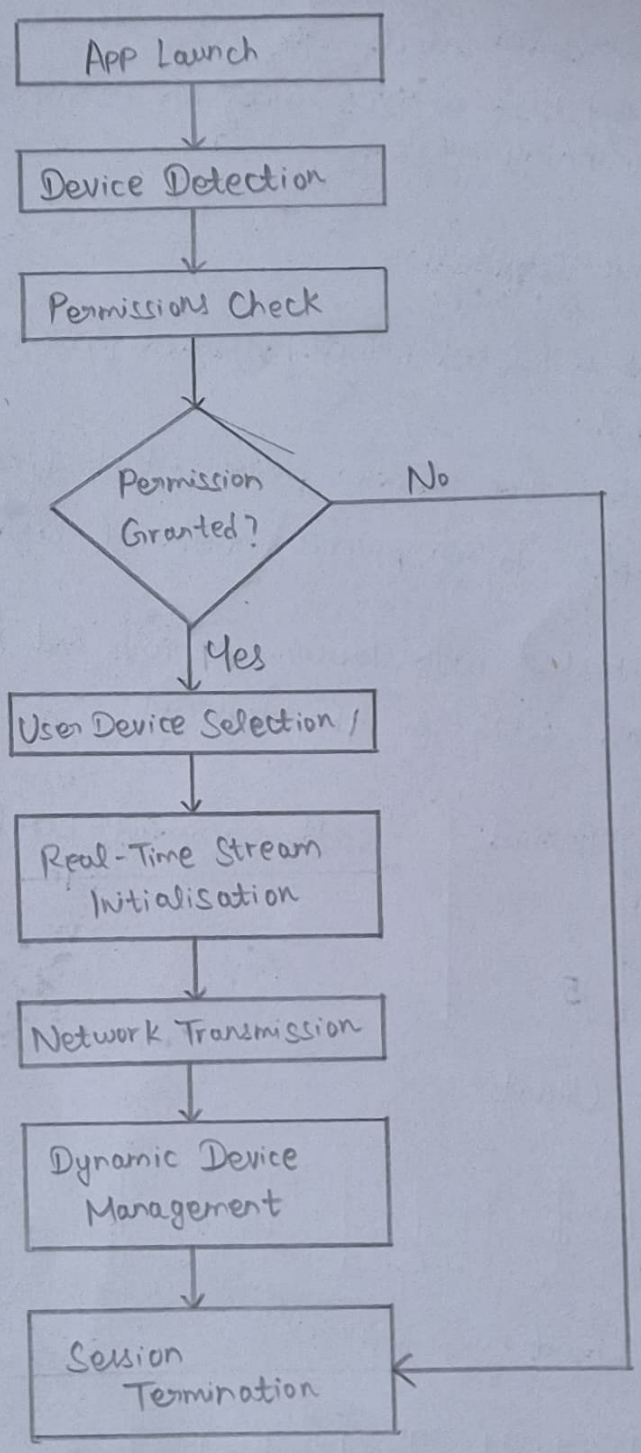
In Adobe Premier Pro, when editing a 4K video, each frame is loaded into memory as you scrub the timeline. The software allocates memory dynamically for effects, transitions, and previews. If memory allocation is inefficient, the preview may lag or crash the application.

Buffering: Preloading data into a reserved memory area to accomodate differences in data processing or transmission speeds between devices or software modules.

During YouTube video playback, a portion of the video is preloaded into a buffer before playing. This helps prevent playback interruption due to momentary drops in internet speed. The larger the buffer, the more seamless the experience.

Illustrate the flow of Multimedia Device Management using a flowchart.

Multimedia Device Management Flow



Briefly explain the basic principles of EDF and RMS scheduling algorithms.

Earliest Deadline First

→ Each task has an executine time and deadline.

→ The scheduler selects the task with the closest deadline.

→ If a new task arrives with an earlier deadline, it preempts the currently running task.

Rate Monotonic Scheduling

→ Assigns a fixed priority → higher priority goes to task with shorter period.

→ Executes the higher priority task first

→ Preempts lower priority tasks if higher-priority ones arrive.

Consider 2 tasks with known periods and ETs. Show the working of EDF and RMS for the task set.

Task	ET	Period
T1	1	2
T2	2	5

Hyperperiod = LCM(2,5)
= 10

RMS Gantt Chart

T1	T2	T1	T2	T1	T2	T1	T2	T1	–

0 1 2 3 4 5 6 7 8 9 10

EDF Gantt Chart

T1	T2	T1	T2	T1	T2	T1	T2	T1	–

0 1 2 3 4 5 6 7 8 9 10

Highlight the key difference between RMS and EDF.

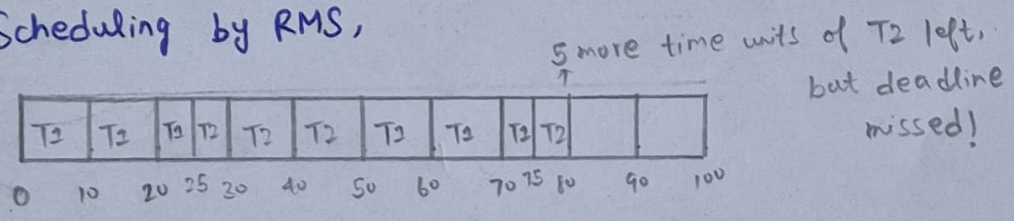
The key difference between RMS and EDF is that RMS assigns a static priority to the task set, whereas EDF assigns a dynamic priority. This means that there are certain task sets with deadlines that RMS fails to schedule, which EDF can schedule.

Give a small example where RMS fails but EDF works.

Consider the given task set.

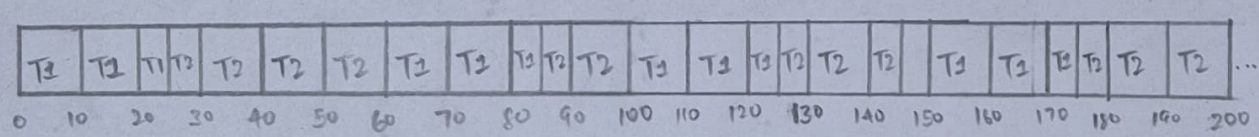
Task	ET	Period
T1	25	50
T2	35	80

Scheduling by RMS,



RMS has thus failed.

Scheduling by EDF,



EDF works.

Which of the two would you recommend for a multimedia system handling tasks with variable deadlines - EDF or RMS? Justify.

EDF is better than RMS for a multimedia system handling tasks with variable deadlines, as it prioritise said deadlines more effectively.